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Effects of physiological integration on defense strategies against herbivory by the clonal plant *Alternanthera philoxeroides*

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Abstract

Aims
The plant–herbivore interaction is one of the most fundamental interactions in nature. Plants are sessile organisms, and consequently rely on particular strategies to avoid or reduce the negative impact of herbivory. Here, we aimed to determine the defense strategies against insect herbivores in the creeping invasive plant *Alternanthera philoxeroides*.

Methods
We tested the defense response of *A. philoxeroides* to herbivory by a leaf-feeding specialist insect *Agasicles hygrophila* and a polyphagous sap-feeding insect *Planococcus minor*. We also tested the mechanisms triggering defense responses of *A. philoxeroides* by including treatments of artificial leaf removal and jasmonic acid application. Furthermore, we examined the effect of physiological integration on these defense strategies.

Important Findings
The combination of artificial leaf removal and jasmonic acid application produced a similar effect to that of leaf-feeding by the real herbivore. Physiological integration influenced the defense strategies of *A. philoxeroides* against herbivores, and increased biomass allocation to aboveground parts in its apical ramets damaged by real herbivores. Our study highlights the importance of physiological integration and modular plasticity for understanding the consequences of herbivory in clonal plants.

Keywords: *Agasicles hygrophila*, alligator weed, clonal integration, herbivory, *Planococcus minor*

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INTRODUCTION

Clonal growth in plants is the production of vegetatively produced modules, named ramets, which can remain physically connected via stolons, rhizomes or horizontally growing roots (Klimes *et al.* 1997; Pitelka and Ashmun 1985; Price and Marshall 1999; Xu *et al.* 2012). One of the most interesting traits associated with clonal plants is the capacity for (clonal) physiological integration that allows the exchange of, e.g. resources and signals between connected ramets of clonal plants (Pitelka and Ashmun 1985). Physiological integration is especially important for developing ramets or ramets located in resource-poor patches, which are supported by resource translocation from established ramets or ramets in resource-rich patches (Alpert 1999; Hartnett and Bazzaz 1983; Jónsdóttir and Watson 1997; Saitoh *et al.* 2002; Roiloa and Retuerto 2006; Slade and Hutchings 1987). Effects of physiological integration have been previously studied in a variety of situations with heterogeneous distribution of environmental factors, including light, nutrients, water, salinity, heavy metals, sand burial, wind erosion, pathogens and defoliation (D'Hertefeldt and van der Putten 1998; Roiloa and Retuerto 2012; Salzman and Parker 1985; Schmid *et al.* 1988; Wang *et al.* 2017a, 2017b; You *et al.* 2014; Yu *et al.* 2004, 2008). Benefits of clonal integration commonly outweigh its potential costs, allowing clonal plants to overcome stressful conditions and colonize a wide array of environments (Roiloa and Retuerto 2012; Roiloa

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et al. 2016; Song et al. 2013; Wang et al. 2017a, 2017b; You et al. 2014).

The plant–herbivore interaction is one of the most fundamental interactions in nature (Begon et al. 2014). Plants are sessile organisms, and consequently rely on particular strategies to avoid or reduce herbivore attacks and/or impacts (Schoonhoven et al. 2005). To study the impact of plant–herbivore interactions, one should also consider the underlying mechanism triggering defense responses in plants. Induced defense mechanisms allow plants injured by herbivores to activate defense responses, whereas such defense responses will not be activated in the absence of herbivory (Agrawal 2000; Karban and Baldwin 1989). Therefore, induced defense has been described as a strategy to maximize the benefits and minimize the costs of defenses in environments where herbivory is variable (Bråthen et al. 2004). Simulated leaf removal is a commonly used approach to study consequences of herbivory. However, plant responses to herbivores are complex and can also be activated by other mechanisms (Baldwin 1990). For instance, the presence of the salivary fluid secretion of herbivores may activate defense responses of plants (Felton and Eichenseer 1999), but leaf removal by itself may not (Baldwin 1990; Felton and Eichenseer 1999). Also, the exogenous application of jasmonic acid or its derivate, methyl jasmonate, may trigger the same defensive responses as those caused by real herbivores (Baldwin 1998; Thaler et al. 1996). Thus, the use of leaf removal and jasmonic acid may help separate mechanical and chemical effects when studying induced defense to herbivory (Agrawal et al. 1999; Baldwin 1996), and the combined application of both treatments may represent a scenario more similar to real herbivory (van Kleunen et al. 2004).

While many studies have examined the effects of clonal integration on responses of clonal plants to both abiotic and biotic factors (Roiloa and Retuerto 2012; Roiloa et al. 2016; Song et al. 2013; Wang et al. 2017b), including those to simulated herbivory (Schmid et al. 1988; Wang et al. 2017a; You et al. 2014), little is known about its effects on the responses of clonal plants to real herbivory (but see Gómez et al. 2007, 2008). Schmid et al. (1988) reported a benefit of clonal integration in defoliated ramets in a simulated herbivory experiment. Similarly, Wang et al. (2017a) found that clonal integration increased the tolerance to heavy defoliation in a rhizomatous plant. The role of clonal integration in generating induced defense to herbivores has also been tested (Gómez and Stuefer 2006; Gómez et al. 2007, 2008). These studies provide evidence that induced systemic resistance (induced defense developed at a non-local scale) is mediated by clonal integration, which can mitigate the damage suffered by defoliated ramets (Gómez and Stuefer 2006; Gómez et al. 2007, 2008). However, no previous study has assessed the mechanisms triggering the defense responses against herbivory and how these responses are affected by clonal integration.

We conducted a greenhouse experiment to test the effect of physiological integration on the defense responses to herbivory in the clonal plant *Alternanthera philoxeroides*. We compared the difference in growth and biochemical responses of *A. philoxeroides* attacked by a leaf-feeding specialist insect *Agasicles hygrophila* and a polyphagous sap-feeding insect, *Planococcus minor*. Also, we examined whether the induced response to herbivory was triggered simply by mechanical damage (leaf removal) or by a combination of mechanical damage and signaling hormones (e.g. jasmonic acid). Finally, we tested the role of clonal integration in the defense strategies of *A. philoxeroides* against herbivores. Specifically, we hypothesized (i) that the leaf-feeding specialist insect will provoke a more negative impact on plant growth than the generalist sap-feeding herbivore, (ii) that the combination of leaf removal and jasmonic acid application will be necessary to activate the defensive response and (iii) that the negative impact of herbivory will be alleviated by physiological integration.

MATERIALS AND METHODS

Study species

Alternanthera philoxeroides (Mart.) Griseb. (Amaranthaceae), commonly known as alligator weed, is an amphibious, perennial herb native to the Parana River basin in South America (Julien et al. 1995). The geographical distribution of the species out of its native range includes the south of North America, Australia, New Zealand, Indonesia, Thailand and China (Julien et al. 1995; Sainty et al. 1998). *Alternanthera philoxeroides* is highly invasive and considered one of the worst weeds in the world (Wang et al. 2008; Yu et al. 2009). This species can grow both on land and in water, and form floating mats in water courses. *A. philoxeroides* in China shows extremely low genetic diversity and does not produce viable seeds (Lu et al. 2013). Instead, it reproduces clonally by stem and/or root fragments (Dong et al. 2010; Julien et al. 1995; Sainty et al. 1998; Yu et al. 2009). It can displace native species, block irrigation systems and increase the risk of floods (Julien and Bourne 1988). Over time, water bodies become swamps covered by the plant (Sainty et al. 1998).

Agasicles hygrophila Selman and Vogt (Coleoptera: Chrysomelidae), commonly known as alligator weed leaf beetle, is a species native to South America (Maddox 1968). The larvae and adult of *A. hygrophila* both feed on leaves and stems of *A. philoxeroides*, and eventually cause its death. The species has been introduced to North America, China and New Zealand as an agent to control *A. philoxeroides*, with excellent results in North America but not always successful in other regions (Burgin et al. 2010; Lu and Ding 2011).

Planococcus minor Maskell (Hemiptera: Pseudococcidae), is a polyphagous pest native to Asia (Cox 1989). Its host range is wide, including more than 250 species in 80 families (Venette and Davis 2004). *Planococcus minor* has a short life cycle, a high

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reproductive rate and a polyphagous nature (Francis *et al.* 2012). The species is considered a pest in India and China, and a potential risk in USA (Venette and Davis 2004).

Experimental design

All plant material used in the experiment was collected in Zhejiang Province, China. A total of 150 clonal fragments, each consisting of six connected ramets and a stem apex, were cut off from stock plants. The two oldest ramets (hereafter called 'basal ramets') of clonal fragments were placed in a container with water to promote root formation. The other four youngest ramets (hereafter called 'apical ramets') remained outside the water. After 15 days, seven groups of plants, each containing 12 clonal fragments of similar size, were selected. Each fragment was placed in two adjacent pots, with the two basal ramets in one pot and the remaining four apical ramets in the adjacent pot. All the pots with apical ramets were enclosed by insect cages (25 cm long × 25 cm wide × 50 cm high).

The experiment consisted of six levels of herbivory crossed with two levels of clonal integration (with or without), making a total of 12 treatments (Fig. 1). There were seven replicates for each treatment. To minimize differences due to possible environmental patchiness in the greenhouse, we used a block design with seven blocks. Each block contained one replicate of each treatment and one group of the 12 plants (clonal fragments) of *A. philoxeroides*. The 12 plants within each group were randomly assigned to the 12 treatments in one block. The six herbivory treatments were a control (no real or simulated herbivory), two treatments of real herbivory (herbivory by the specialist herbivore *A. hygrophila* and herbivory by the generalist *P. minor*) and three treatments of simulated herbivory (jasmonic acid application, artificial leaf removal and both jasmonic acid application and artificial leaf removal). For the treatments without clonal integration, the apical and basal parts of a clonal fragment were disconnected by cutting off the stem internode connecting them, and for the treatments with clonal integration, they were left connected.

We released four adults of *A. hygrophila* into each cage in the specialist herbivory treatment, and put 15–25 adults of *P. minor* on the stems of the apical ramets in each cage in the generalist herbivory treatment. In the leaf removal treatment, we artificially removed 50% of the leaves that were over 2 cm long in the apical ramets. This treatment setup has been previously described as an accurate mean of simulated herbivory damage of *A. philoxeroides* by *A. hygrophila* (Schooler *et al.* 2006). In the jasmonic acid application treatment, we sprayed the apical ramets with a 100 μmol L^{-1} jasmonic acid solution (J2500-100MG; Sigma-Aldritch, St.

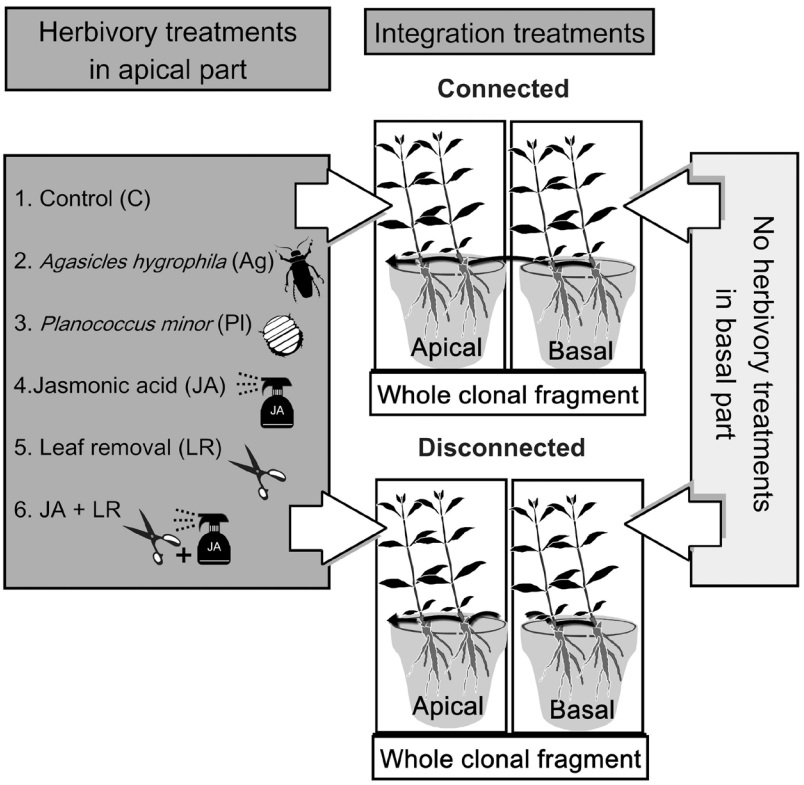

Figure 1: schematic representation of the experimental treatments consisting of two crossed factors with herbivory (control, leaf-feeding *Agasicles hygrophila*, sap-feeding *Planococcus minor*, jasmonic acid, leaf removal, jasmonic acid + leaf removal) and clonal integration (connected, disconnected) as main factors.

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Louis, MO, USA) with a 1:1 mixture of 95% ethanol and distilled water as the solvent. The target ramets in the jasmonic acid application treatment were covered by a plastic bag until the solution dried out to prevent it from affecting neighbor plants due to its volatility. In the treatment with both jasmonic acid application and leaf removal, we removed 50% of the leaves of the apical ramets, which were over 2 cm long and then sprayed them with the jasmonic acid solution. Leaf removal and jasmonic acid application were treated once a week.

The experiment was carried out in a greenhouse belonging to Forest Science Company, Ltd., of Beijing Forestry University in Beijing, China. Plants were grown under a natural photoperiod and watered regularly to avoid water stress. During the experiment, the air temperature in the greenhouse was 27.5 ± 0.3°C, and the relative humidity 81.3 ± 1.5% (mean ± SE), measured daily with Hobo Temp/RH loggers (HOBO UX100-003; Onset Computer Co., Bourne, MA, USA). Treatments began on 10 July 2016, and continued for 5 weeks.

Measurements

At harvest, number of ramets of the apical part and the basal part were counted separately. The apical part and basal part were each divided into leaves, stems and roots, dried at 75°C for 48 h and then weighed.

We also measured concentrations of total phenolics and condensed tannins of leaf samples. The method for determining total phenolics using Folin Ciocalteu reagent was adapted from Mcdonald *et al.* (2001); the method for measuring tannins was the acid butanol assay described by Gessner and Steiner (2005). A total of 50 mg of dried, ground leaf material was needed for each measurement. For lack of sufficient leaf material, only five replicates were used for these measurements.

Data analysis

We used three-way analysis of variance (ANOVA) to test effects of block, clonal integration, herbivory and clonal integration × herbivory on total mass, leaf mass, stem mass, root mass, number of ramets and root to shoot ratio of the apical part, the basal part and the clonal fragment (apical plus basal part). We used two-way ANOVA to examine effects of clonal integration, herbivory and their interaction on total phenolics and tannins. All data were checked for normality and homoscedasticity. Where necessary, they were transformed to meet the requirements of ANOVA. As a consequence, the square root transformation was applied to the following variables: leaf mass, root mass and root to shoot ratio of the clonal fragment, stem mass, root to shoot ratio and tannins of the apical part, and number of ramets, leaf mass, root mass, stem mass and root to shoot ratio of the basal part. When the effects were significant, we applied a *posteriori* Tukey test to detect differences among the six herbivory treatments, and *t*-test for differences between the connected and disconnected treatments within each herbivory treatment.

Mortality during the experiment affected one apical part from the disconnected treatment with herbivory by *A. hygrophila*. Two of the basal parts did not develop, i.e. one from the connected, control treatment and the other from the disconnected treatment with herbivory by *A. hygrophila*. Dead or undeveloped plants were not included in the analyses. Significance levels were set at $P = 0.05$ after Bonferroni correction. All analyses were conducted with IBM SPSS, version 23 (IBM Corp., Armonk, NY, USA).

RESULTS

Performance of basal ramets

Connection with the apical ramets significantly increased both root mass and root to shoot ratio of the basal ramets when the apical ramets were subjected to herbivory by *A. hygrophila*, and also increased root to shoot ratio of the basal ramets when the apical ramets were subjected to jasmonic acid application plus leaf removal (Fig. 2e and f, Table 1). Connection significantly reduced the tannin concentration in the basal ramets when the apical ramets were treated with jasmonic acid (Fig. 4, Table 2). However, connection had no significant effect on other growth estimates or on the concentration of phenolics of the basal ramets (Tables 1 and 2, Figs 2 and 4). In addition, herbivory of the apical ramets had no significant effect on the growth or physiology of the basal ramets (Tables 1 and 2, Figs 2 and 4).

Performance of apical ramets

Herbivory significantly affected all growth measures and root to shoot ratio of the apical ramets (Table 1). Compared to the control, herbivory by *A. hygrophila*, leaf removal and jasmonic acid application plus leaf removal generally reduced biomass of the apical ramets (Fig. 3a–e). Connection with the basal ramets significantly decreased root to shoot ratio of the apical ramets, especially when the apical ramets were attacked by *A. hygrophila* and *P. minor* (Fig. 3f). Herbivory of the apical ramets significantly affected the tannin concentration of the apical ramets (Table 2). The apical ramets with jasmonic acid application showed the lowest tannin concentration (Fig. 4d).

Performance of clonal fragments

Herbivory of the apical ramets significantly affected leaf, stem, root and total mass of the whole clonal fragment (apical plus basal ramets; Table 1). Compared to the control, biomass of the clonal fragment reduced when the apical ramets were subjected to herbivory by *A. hygrophila*, leaf removal and jasmonic acid application plus leaf removal (Fig. 5a, c–e). Root to shoot ratio of the clonal fragment was also affected by herbivory (Table 1), showing the highest value when the apical ramets were attacked by the specialist herbivore *A. hygrophila* (Fig. 5f).

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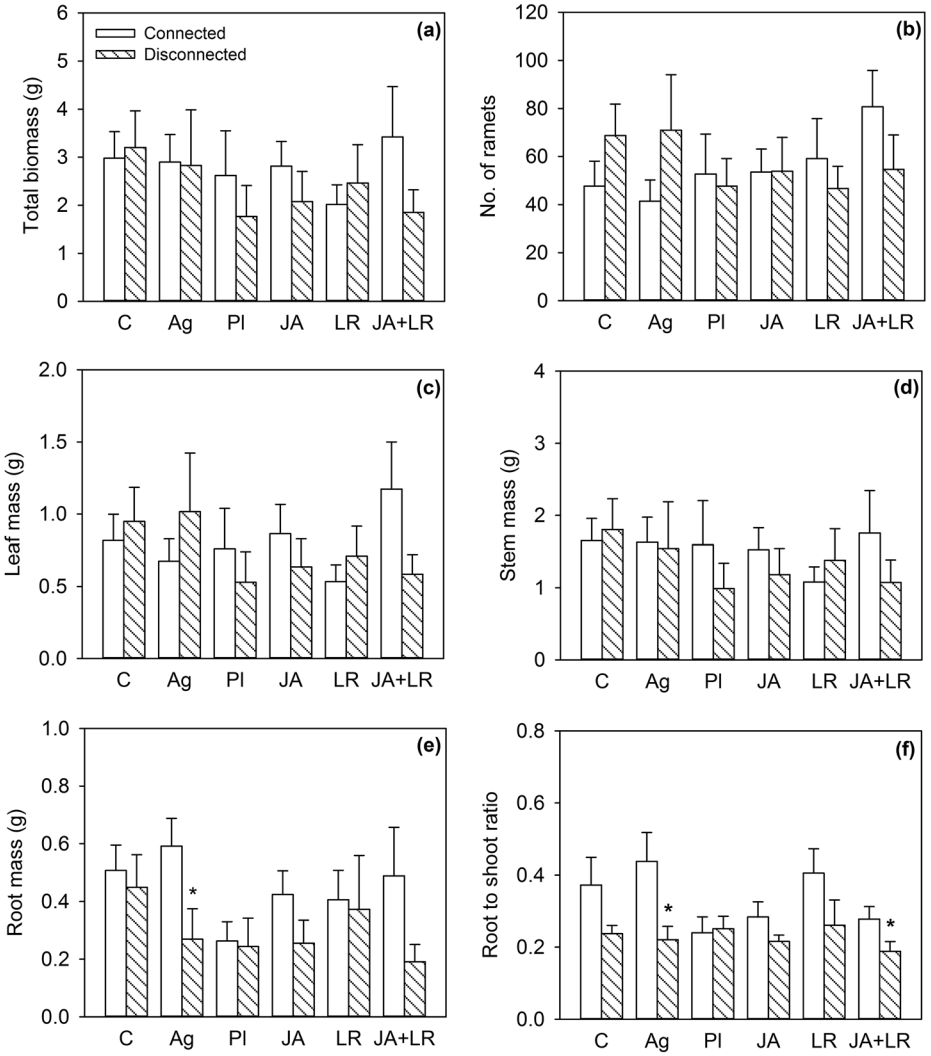

Figure 2: biomass (a, c–e), number of ramets (b) and root to shoot ratio (f) of the basal ramets of *Alternanthera philoxeroides*. Treatment codes: C—control, Ag—herbivory by the specialist *Agasicles hygrophila*, Pl—herbivory by the generalist *Planococcus minor*, JA—jasmonic acid application, LR—leaf removal and JA+LR—jasmonic acid application plus leaf removal. Values are mean + SE. Stars indicate significant differences between the connection and disconnection treatment within each herbivory treatment. See Table 1 for ANOVA results.

DISCUSSION

We compared the growth and physiological responses of the invasive plant *A. philoxeroides* between two types of aboveground herbivory, i.e. by a leaf-feeding specialist *A. hygrophila* and a polyphagous sap-feeding herbivore *P. minor*. As we predicted, the negative impact on apical ramets was stronger when *A. philoxeroides* was consumed by *A. hygrophila* than by *P. minor*. This is because the two insect species caused different kinds of damage to the host plant. Both adults and larvae of *A. hygrophila* devoured the leaves and external part of the stems, resulting in lower total, leaf and stem mass compared to the control. Damage was smaller in plants attacked by *P. minor* whose feeding system sucks the phloem. The reduction of photosynthetic structures caused by the leaf-feeding insect will reduce the net carbon gain of *A. philoxeroides*, as its photosynthetic capacity will be reduced but respiration in other tissues will be maintained. On the other hand, the sap-feeding insect is not expected to damage photosynthetic structures so that plant growth will be less affected.

Our second objective was to determine the mechanism triggering defense responses of *A. philoxeroides* to herbivory. For morphometric variables of apical ramets, the effects of leaf removal and the combined application of jasmonic acid and leaf removal were similar to the effects caused by the real attack of the specialist herbivore *A. hygrophila*. These results are reasonable because *A. hygrophila* is a leaf-feeding herbivore and mechanical leaf removal has been described as an accurate simulation of its herbivory damage in *A. philoxeroides* (Schooler *et al.* 2006). Interestingly, we

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Table 1: ANOVAs results for effects of block, herbivory treatments, connection (clonal integration) and herbivory × connection (H × C) on growth and root to shoot ratio of the basal part, the apical part and the whole clonal fragment of *Alternanthera philoxeroides*

	Basal part			Apical part			Clonal fragment		
	df	*F*	*P*	df	*F*	*P*	df	*F*	*P*
Total mass									
Block	6	21.05	0.066	6	5.09	0.027	6	13.31	**0.015**
Herbivory	5	0.87	0.511	5	10.84	**<0.001**	5	10.12	**<0.001**
Connection	1	1.91	0.216	1	1.27	0.303	1	3.02	0.133
H × C	5	0.98	0.446	5	0.88	0.505	5	1.61	0.190
Residuals	64			66			63		
No. of ramets									
Block	6	2.64	0.125	6	5.96	0.023	6	3.56	0.097
Herbivory	5	0.51	0.769	5	3.31	**0.017**	5	1.80	0.143
Connection	1	<0.01	0.981	1	1.93	0.214	1	0.29	0.609
H × C	5	0.95	0.466	5	0.96	0.457	5	1.92	0.123
Residuals	64			66			63		
Leaf mass									
Block	6	14.48	0.025	6	4.4	0.055	6	9.72	0.098
Herbivory	5	1.06	0.403	5	10.58	**<0.001**	5	9.11	**<0.001**
Connection	1	0.95	0.367	1	2.57	0.160	1	2.63	0.156
H × C	5	1.36	0.270	5	0.60	0.700	5	1.86	0.133
Residuals	64			66			63		
Stem mass									
Block	6	17.27	**0.013**	6	4.15	0.043	6	9.27	**0.016**
Herbivory	5	0.78	0.569	5	9.18	**<0.001**	5	9.91	**<0.001**
Connection	1	4.12	0.089	1	1.58	0.255	1	2.71	0.151
H × C	5	0.73	0.606	5	1.02	0.424	5	1.33	0.280
Residuals	64			66			63		
Root mass									
Block	6	6.12	0.025	6	34.39	0.399	6	26.02	0.132
Herbivory	5	2.17	0.085	5	4.45	**0.004**	5	3.30	**0.017**
Connection	1	9.20	0.023	1	5.47	0.058	1	0.49	0.511
H × C	5	1.39	0.259	5	1.86	0.131	5	2.33	0.069
Residuals	64			66			63		
Root to shoot ratio									
Block	6	1.00	0.469	6	4.50	0.131	6	8.92	0.171
Herbivory	5	1.13	0.367	5	3.14	**0.021**	5	3.45	**0.014**
Connection	1	11.88	**0.014**	1	21.29	**0.004**	1	3.41	0.114
H × C	5	2.33	0.069	5	1.92	0.121	5	1.58	0.198
Residuals	64			66			63		

F and *P* values are given. Values for which $P < 0.05$ (after Bonferroni correction) are given in bold. See Figs 2, 3, and 5 for data.

detected a non-local effect of herbivory on basal ramets. In a compensatory response, basal ramets connected to apical ramets under *A. hygrophila* herbivory and the combined application of jasmonic acid and leaf removal significantly increased root biomass of basal ramets. The absence of a similar compensatory response under the attack by the sap-feeding generalist *P. minor* suggests that leaf damage is necessary to trigger this non-local response. Our results also indicate that a combination of mechanical leaf removal and jasmonic acid application is required to induce this compensatory response, whereas leaf removal alone may not have an appreciable effect. Such results are in line with previous studies reporting that the combined use of leaf removal and jasmonic acid may represent a more realistic scenario to mimic real herbivory (van Kleunen *et al.* 2004).

Several studies have showed that the exogenous application of jasmonic acid or its derivate provokes the same defensive responses in different plant species as those caused by

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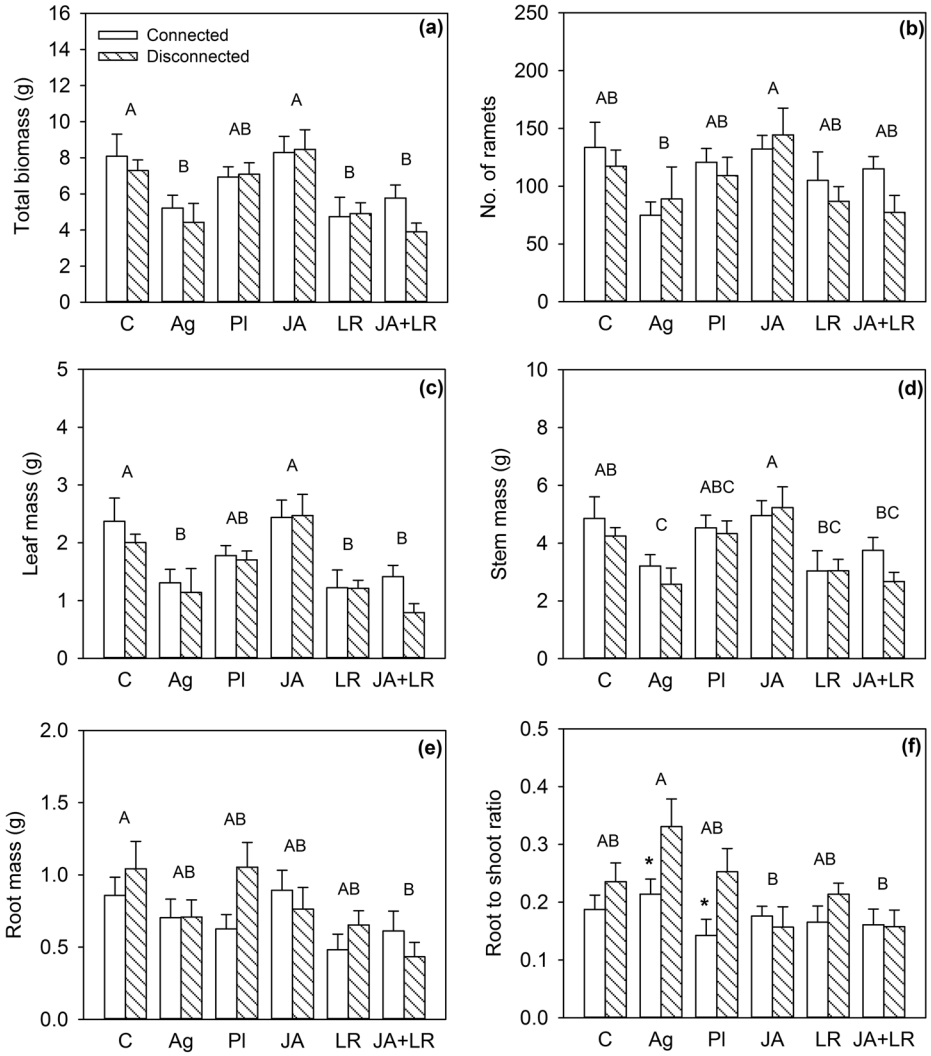

Figure 3: biomass (a, c–e), number of ramets (b) and root to shoot ratio (f) of the apical ramets of *Alternanthera philoxeroides*. Values are mean + SE. Letters on the bars indicate differences between the herbivory treatments. Stars indicate significant differences between the connection and disconnection treatment within each herbivory treatment. See Fig. 2 for treatment codes and Table 1 for ANOVA results.

predators (Baldwin 1998; Thaler *et al.* 1996). Several members of the jasmonate family have been identified as intermediates in the long-range defensive signaling (Huot *et al.* 2014; Truman *et al.* 2007). Moreover, a number of studies have detected the role of jasmonic acid in defensive responses, including the generation of chemical compounds (Baldwin 1998; Cipollini and Sipe 2001; Thaler *et al.* 1996), the alteration of carbon allocation (Babst *et al.* 2005, Henkes *et al.* 2008) and long-distance defensive signaling (Huot *et al.* 2014; Truman *et al.* 2007).

Contents of phenolic compounds (including tannins) in leaves are often used as a measure of non-specific, quantitative and carbon-based anti-herbivory defense of plants (Asquith and Butler 1986; Forkner *et al.* 2004). These compounds can lead to the inhibition of enzymes (e.g. in the digestive tract of insects) or the formation of the insoluble complex with plant proteins, reducing their dietary value (Asquith and Butler 1986). Several studies have reported a correlation between the tannin or total phenolic content and the negative effect on feeding herbivores (Forkner *et al.* 2004; Karowe 1989). However, we did not find significant differences in the contents of phenolics and tannins between the control and the real or simulated herbivory treatments. A plausible explanation for this observation is that *A. philoxeroides* may have used constitutive defense that are developed permanently (Wittstock and Gershenzon 2002), regardless of the presence or the type of herbivory. As a result, induced chemical defense was not activated during our experiment. It appears that *A. philoxeroides* based its defensive strategy against herbivory on compensatory growth responses, rather than investing resources in the synthesis of defensive chemicals. By reducing the investment in costly chemical defensive

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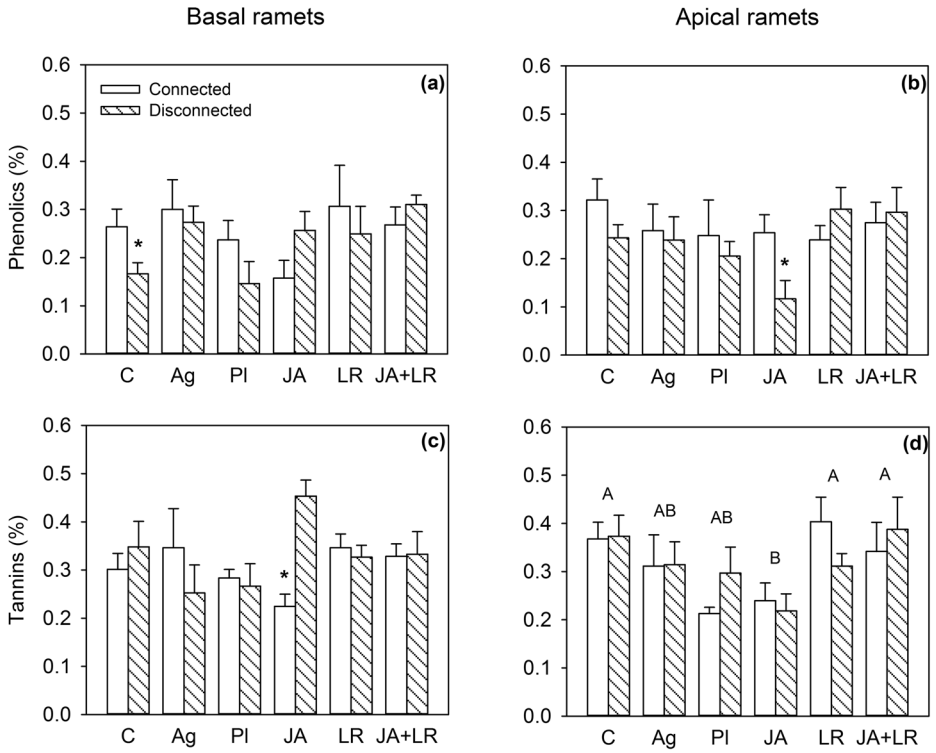

Figure 4: concentrations of total phenolics (a, b) and condensed tannins (c, d) in basal and apical ramets of *Alternanthera philoxeroides*. Values are mean + SE. Letters on the bars indicate differences between the herbivory treatments. Stars indicate significant differences between the connection and disconnection treatment within each herbivory treatment. See Fig. 2 for treatment codes and Table 2 for ANOVA results.

Table 2: ANOVA results for effects of herbivory, connection (clonal integration) and herbivory × connection (H × C) on the total phenolics and condensed tannins in basal and apical ramets of *Alternanthera philoxeroides*

	Basal part			Apical part		
	df	*F*	*P*	df	*F*	*P*
Phenolics						
Herbivory	5	1.87	0.118	5	1.47	0.217
Connection	1	0.66	0.420	1	1.51	0.225
H × C	5	1.42	0.233	5	1.25	0.300
Residuals	48			48		
Tannins						
Herbivory	5	0.69	0.636	5	3.74	**0.006**
Connection	1	1.01	0.320	1	0.04	0.836
H × C	5	3.24	**0.013**	5	0.80	0.556
Residuals	48			48		

F and *P* values are given. Values for which $P < 0.05$ (after Bonferroni correction) are given in bold. See Fig. 4 for data.

compounds, *A. philoxeroides* could spend more resources on growth, thus obtaining a competitive advantage over other species.

Our third objective was to test whether physiological integration can help buffer the negative impact of herbivory. Interestingly, physiological integration significantly reduced root to shoot ratio and thus increased proportional biomass allocation to aboveground structures of apical ramets under real herbivory by both insects. Increasing biomass allocation to aboveground parts could potentially increase the relative ability of *A. philoxeroides* to expand laterally. Therefore, the plastic response mediated by physiological integration could be considered a resistance strategy against herbivores, and might contribute to the successful expansion of *A. philoxeroides*. However, clonal integration was found to increase biomass allocation to shoots of defoliated apical ramets of *A. philoxeroides* (You *et al.* 2014). There are at least two plausible explanations for such a discrepancy. First, our response was detected under real herbivory, while You *et al* (2014) only simulated herbivory by leaf removal. Second, You *et al.* (2014) applied the leaf removal treatment only at the beginning of the experiment, while our insect herbivory was continuous throughout the experiment. These discrepancies call attention to the importance of considering the type (real vs. simulated) and duration (continuous vs. occasional) of the treatments when testing herbivory effects.

Physiological integration also affected biomass allocation of basal ramets, and basal ramets significantly increased biomass allocation to roots when they were connected to apical ramets consumed by the leaf-feeding herbivore *A. hygrophila*. This result could be interpreted as a non-local compensatory

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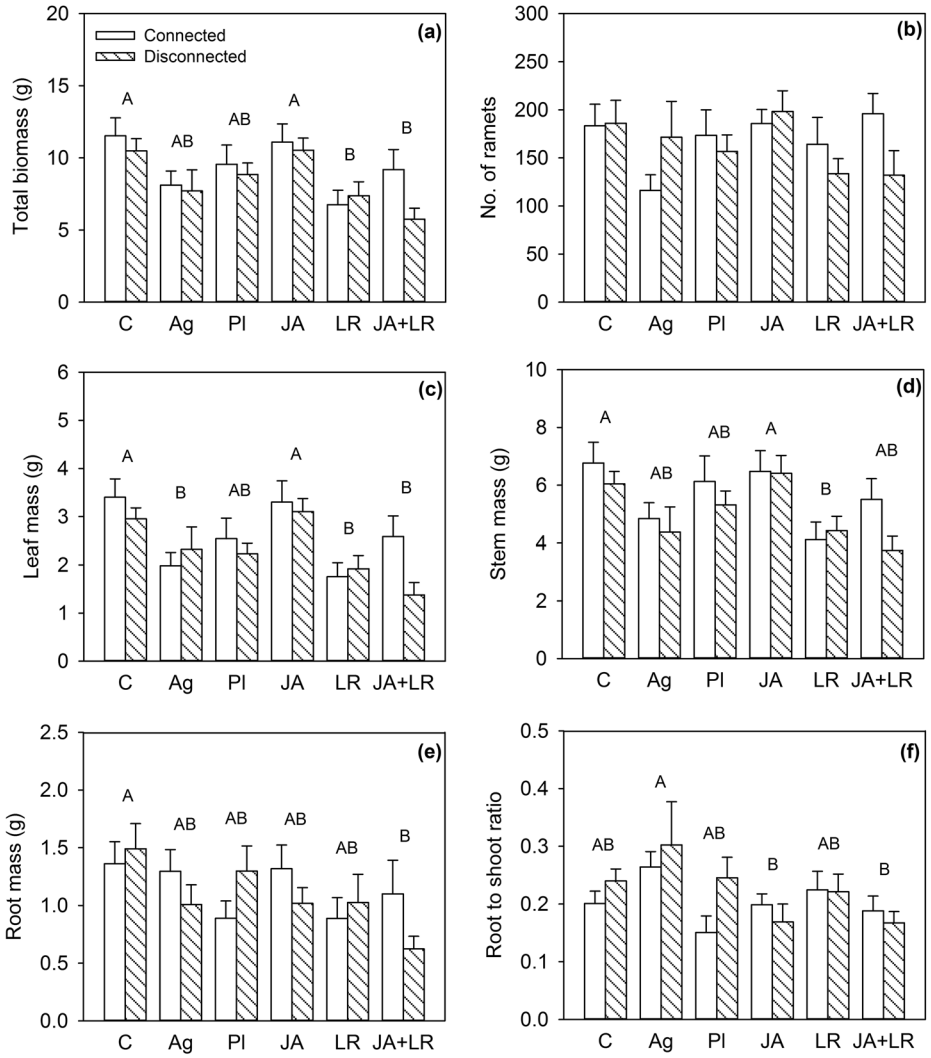

Figure 5: biomass (a, c–e), number of ramets (b) and root to shoot ratio (f) of the complete clonal system of *Alternanthera philoxeroides*. Values are mean + SE. Letters on the bars indicate differences between the herbivory treatments. See Fig. 2 for treatment codes and Table 1 for ANOVA results.

response of basal ramets to meet the demand of apical ramets under the stressful conditions created by herbivory. The plastic response detected in basal ramets in response to local conditions experienced by apical ramets agrees with the modular concept of phenotypic plasticity in plants, which proposes that physiological integration can modify local responses of modules (de Kroon *et al.* 2009). It is important to note that the non-local compensatory response of basal ramets mediated by physiological integration was only detected when apical ramets were attacked by the leaf-feeding herbivore. As discussed above, the leaf-feeding insect *A. hygrophila* imposed the strongest negative effects on apical ramets and, therefore, the compensatory response was only activated under this attack.

Previous studies have reported that physiological integration can induce systemic resistance within a clone (Gómez and Stuefer 2006; Gómez *et al.* 2007, 2008). Consequently, un-attacked ramets can also activate defense in responses to the damage suffered by other members of the clone, reducing the potential negative impact of herbivory on the whole clone. However, we did not detect induced systemic resistance in basal ramets that remained connected to apical ramets suffering from real or simulated herbivory. The alerting signal is probably transmitted by phloem following a source-sink gradient of resources (Gómez and Stuefer 2006). In our experiment, the predominant gradient of resources and thus the direction of resource translocation were from un-attacked basal ramets to apical ramets suffering from herbivory or leaf removal. As there was no predominant gradient of resources from apical to basal ramets, the alerting signal could not be transported from apical to basal ramets so that induced systemic resistance could not occur (Gómez and Stuefer 2006).

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We conclude that physiological integration influences the defense strategies of *A. philoxeroides* against herbivores. Physiological integration increased the allocation to aboveground parts in apical ramets attacked by herbivores, resulting in a more extensive lateral growth. Our results highlight the importance of physiological integration and modular plasticity for the interpretation of the effect of herbivory in clonal plants. In addition, the combination of leaf removal and jasmonic acid application plays a similar role in triggering the compensatory response of *A. philoxeroides* to that of herbivory by the real leaf-feeding insect. Differences between leaf-feeding and sap-feeding herbivores and between real and simulated herbivory should be taken into account to disentangle the defensive response of clonal plants.

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